SDA MINI-PROJECT STUDY GUIDE

Statistical Analysis of Autonomous Vehicle Radar Telemetry

A Student-Friendly Guide: From Radar Physics to Python Data Analytics and Fault Detection

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Course: Statistical Data Analytics (SDA)

Class: SY B.Tech Electronics & Computer Engineering

Academic Year: 2026–27 | Semester I

Project Notebook: `autonomous_radar_telemetry_analysis.ipynb`

Dataset: `autonomous_radar_telemetry_50k.csv` (50,000 observations)

1. Introduction: Autonomous Driving and Radar Sensors

1.1 How Self-Driving Cars See

Self-driving cars cannot rely on just human eyes. They use three main sensors:

- **Cameras:** Great at reading lane lines, speed signs, and traffic light colors, but struggle in darkness, glare, or heavy rain.
- **LiDAR:** Fires thousands of infrared laser beams to create a 3D point cloud of surroundings, but laser pulses reflect off raindrops and fog.
- **mmWave Radar:** Transmits 77 GHz radio waves. Radio waves pass straight through rain, fog, and dust. Radar also gives instant speed measurements using the Doppler Effect.

1.2 What is Sensor Fusion?

Sensor Fusion means combining inputs from Camera, LiDAR, and Radar so the car's computer has a single, reliable picture of the road.

However, if the radar gets damaged or knocked out of alignment, it feeds wrong data into the fusion system. Our project analyzes radar telemetry to detect when a sensor has degraded.

1.3 The 20 Features in Our Dataset

Each row in `autonomous_radar_telemetry_50k.csv` is a snapshot from the radar:

Category	Features	Simple Meaning
Context	`frame_id`, `vehicle_speed_kmh`	Snapshot ID and our car's speed (0 to 130 km/h).
Radar Physics	`radar_frequency_ghz`, `tx_power_dbm`, `rx_gain_db`, `snr_db`, `radar_rcs_dbm2`	Operating frequency (~77.5 GHz), transmit power (~12 dBm), receiver gain (~35 dB), and signal-to-noise ratio.
Target Details	`target_range_meters`, `doppler_velocity_mps`, `azimuth_angle_deg`, `elevation_angle_deg`	Obstacle distance (1 to 200 m), relative speed (+/- 45 m/s), horizontal and vertical angles.
Hardware Health	`sensor_temp_c`, `mcu_voltage_v`, `can_bus_load_pct`, `packet_latency_ms`, `calibration_drift_index`	Temperature (~45°C), MCU power rail (3.30 V), network load, latency (~15 ms), and calibration error (0 to 1).
Environment	`ambient_lux`, `precipitation_rate_mmh`, `lidar_point_density`	Sunlight brightness (0 to 100k lux), rain rate (mm/h), LiDAR point density.
Target Flag	`sensor_degraded_flag`	**0 = Healthy** (78.3%), **1 = Degraded** (21.7%).

2. Exploratory Data Analysis (EDA)

2.1 Checking Sensor Health Distribution

In Python, we count how many sensors are healthy vs degraded:

```
health_counts = df['sensor_degraded_flag'].value_counts()
print(health_counts)
```

Out of 50,000 frames:

- **Healthy (0):** 39,167 frames (78.3%)
- **Degraded (1):** 10,833 frames (21.7%)

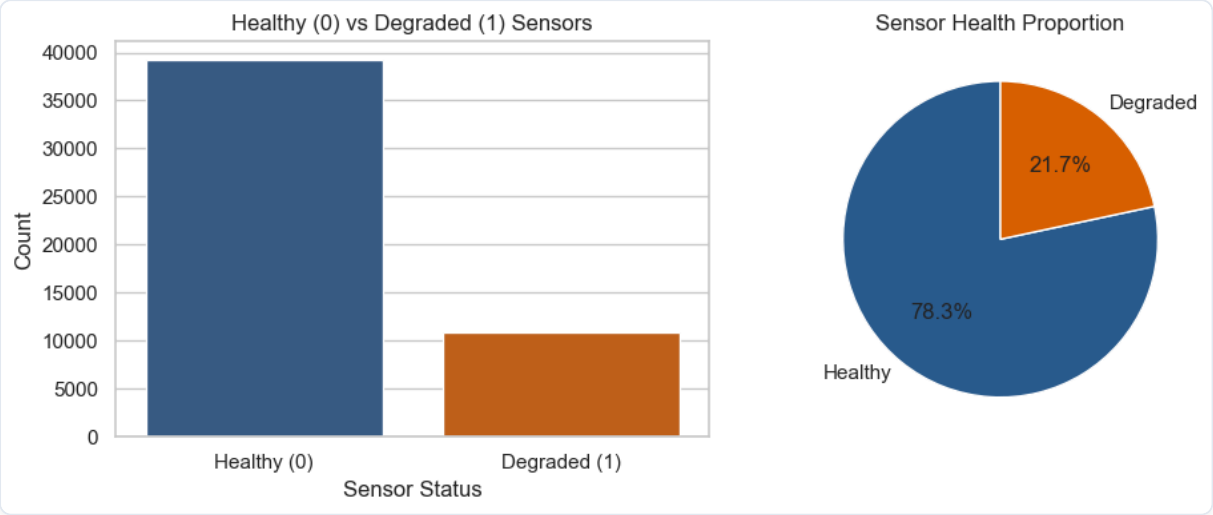


Figure 2.1: Bar plot and Pie chart of healthy vs degraded sensors.

2.2 Distribution of Key Telemetry Parameters

We plot histograms with KDE curves to inspect the spread of key variables:

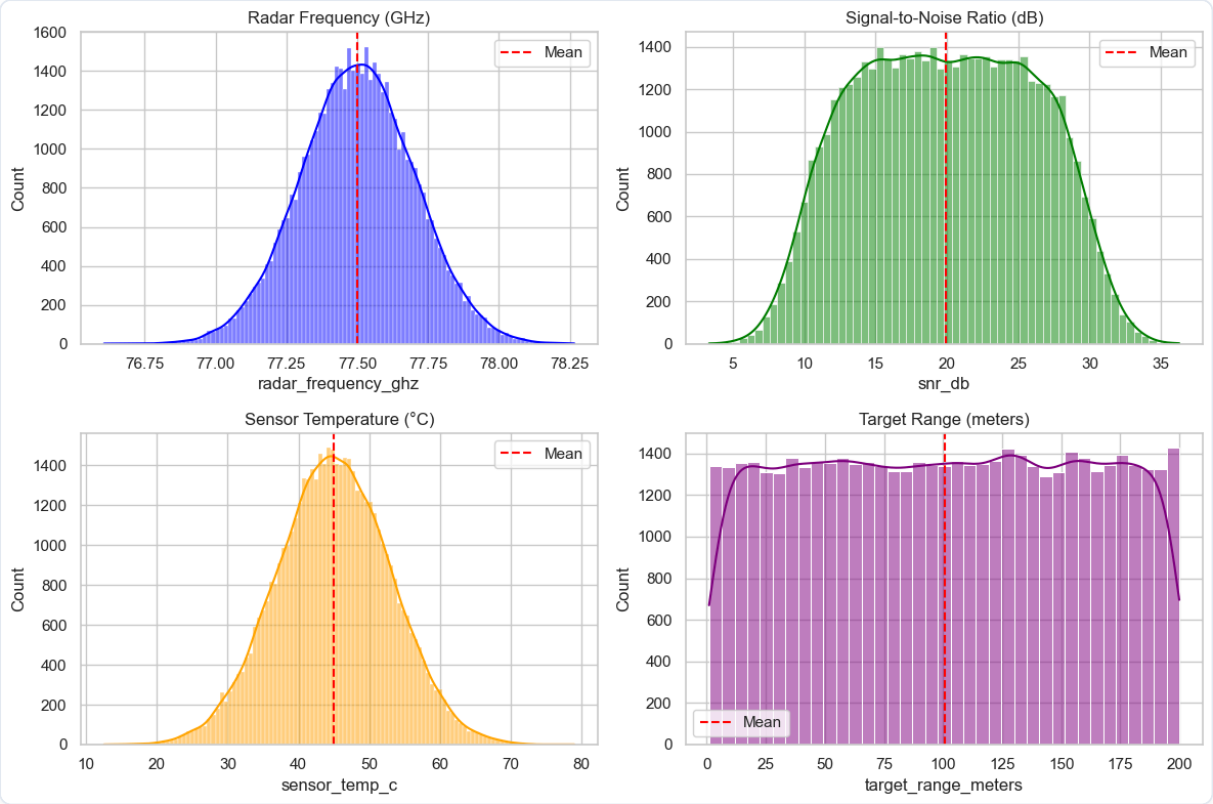


Figure 2.2: Histograms for radar frequency, received SNR, sensor temperature, and target range.

- **Radar Frequency:** Centered at 77.5 GHz like a textbook normal distribution.
- **Sensor Temperature:** Most frames run around 45°C.
- **Target Range:** Spread evenly between 0 and 200 meters.

2.3 Correlation Matrix

We compute the correlation matrix using `df.corr()` to see which features are related:

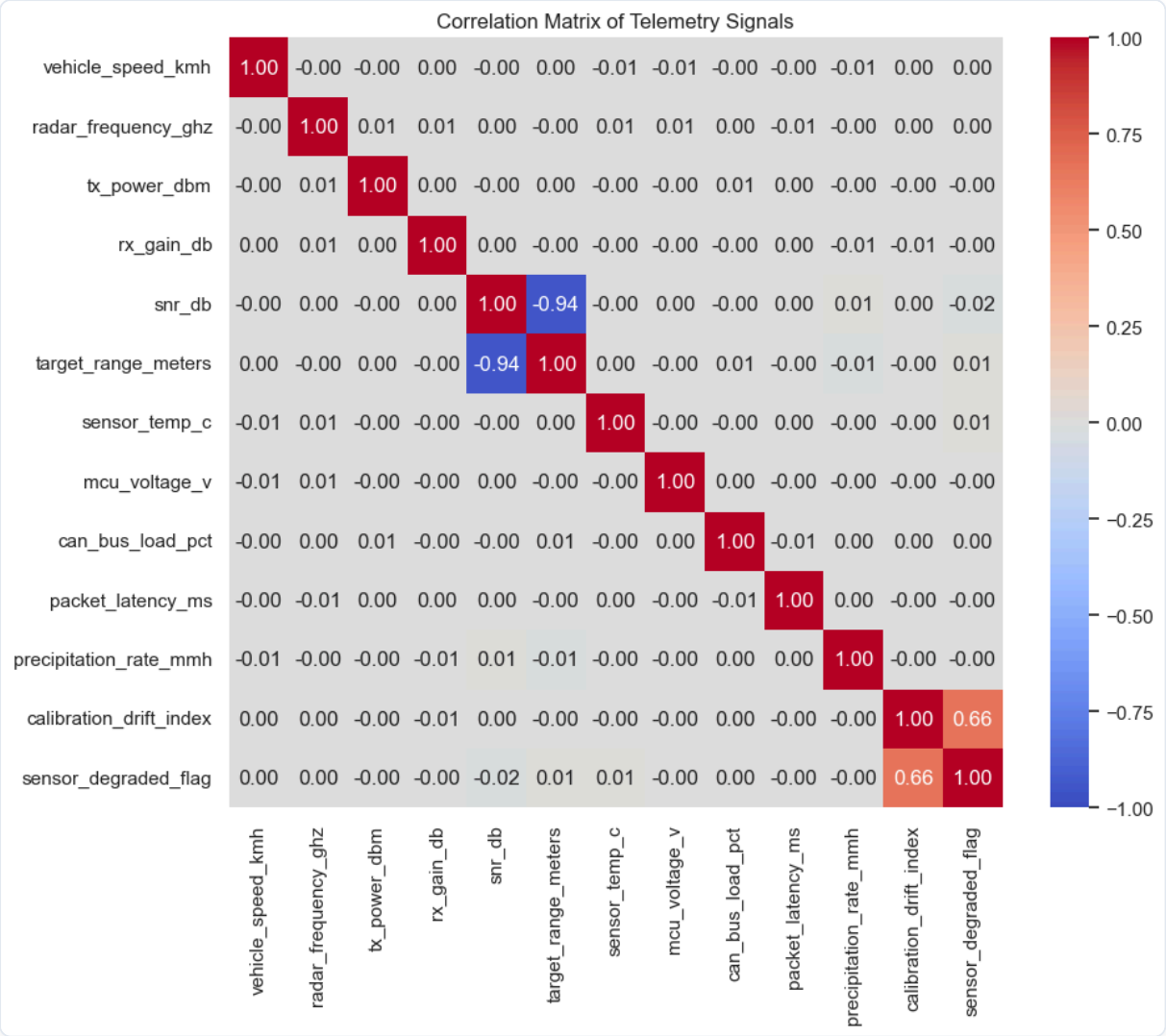


Figure 2.3: Heatmap showing correlation coefficients between features.

3. Solving the Proposed Statistical Investigations

3.1 Question 1: How does target distance influence received SNR?

In physics, radio signals become weaker as they travel through the air. In our notebook, we plot distance vs SNR and run a simple linear regression:

```
slope, intercept, r_value, p_value, std_err = stats.linregress(df['target_range_meters'], df['snr_db
print(f"Slope: {slope:.4f} dB/m")
print(f"R-squared: {r_value**2:.4f}")
```

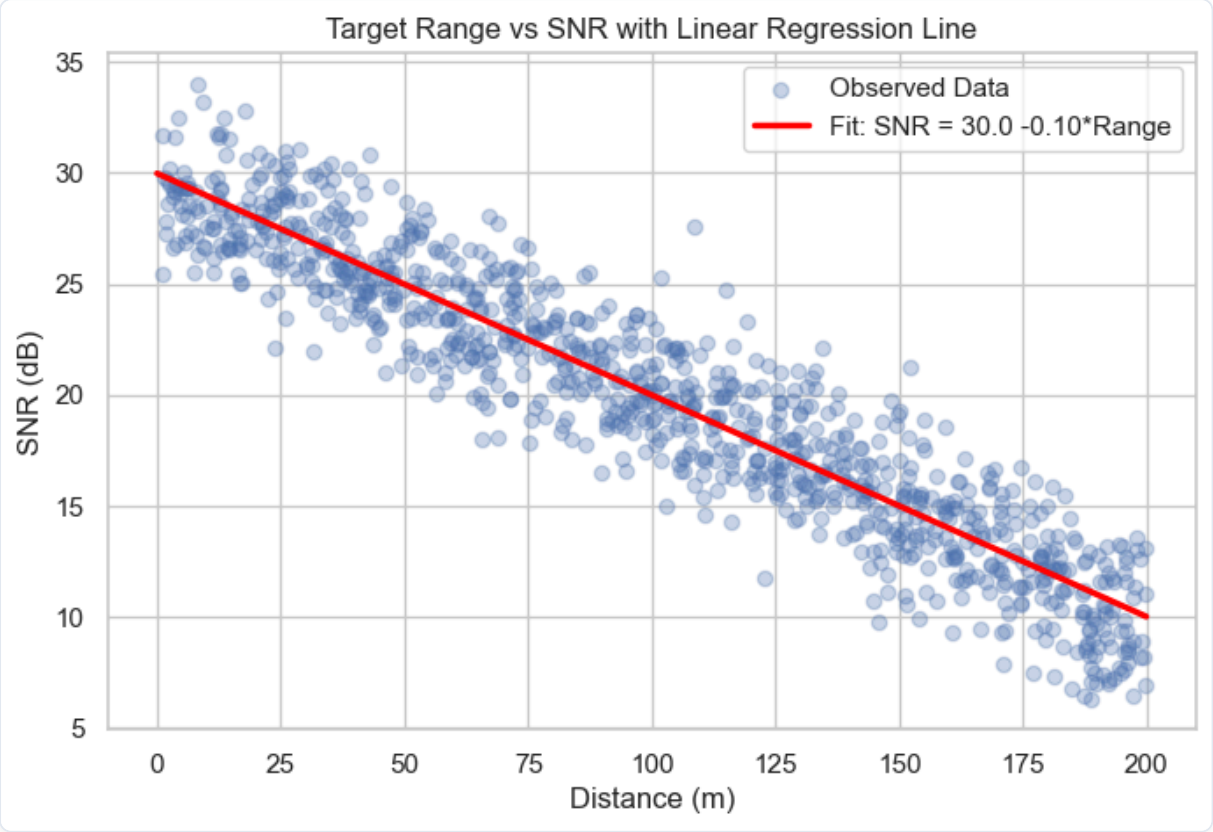


Figure 3.1: Scatter plot and fitted regression line of Target Range vs SNR.

Key Finding for Question 1:

- **Correlation (r):** -0.94 (very strong negative correlation).
- **Slope:** -0.10 dB/meter. For every 10 meters the obstacle moves farther away, received signal power drops by 1 dB.
- **R-squared (R^2):** 0.89. Distance alone explains 89% of the variation in SNR!

3.2 Question 2: Does precipitation affect LiDAR point density?

```
r_rain, p_rain = stats.pearsonr(df['precipitation_rate_mmh'], df['lidar_point_density'])
```

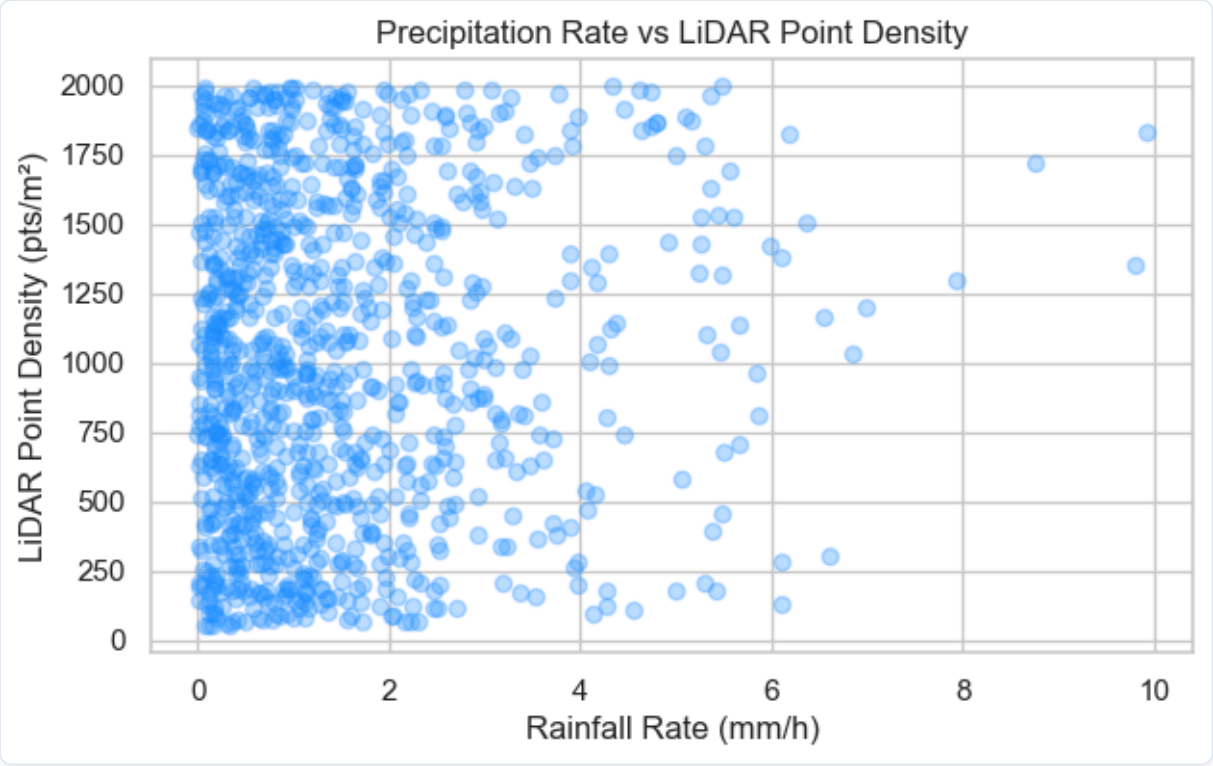


Figure 3.2: Rainfall rate vs LiDAR point density.

The correlation is essentially 0.00 ($p > 0.05$), meaning rainfall rate has no linear association with raw point count in this dataset.

3.3 Question 3: Does CAN bus load affect communication latency?

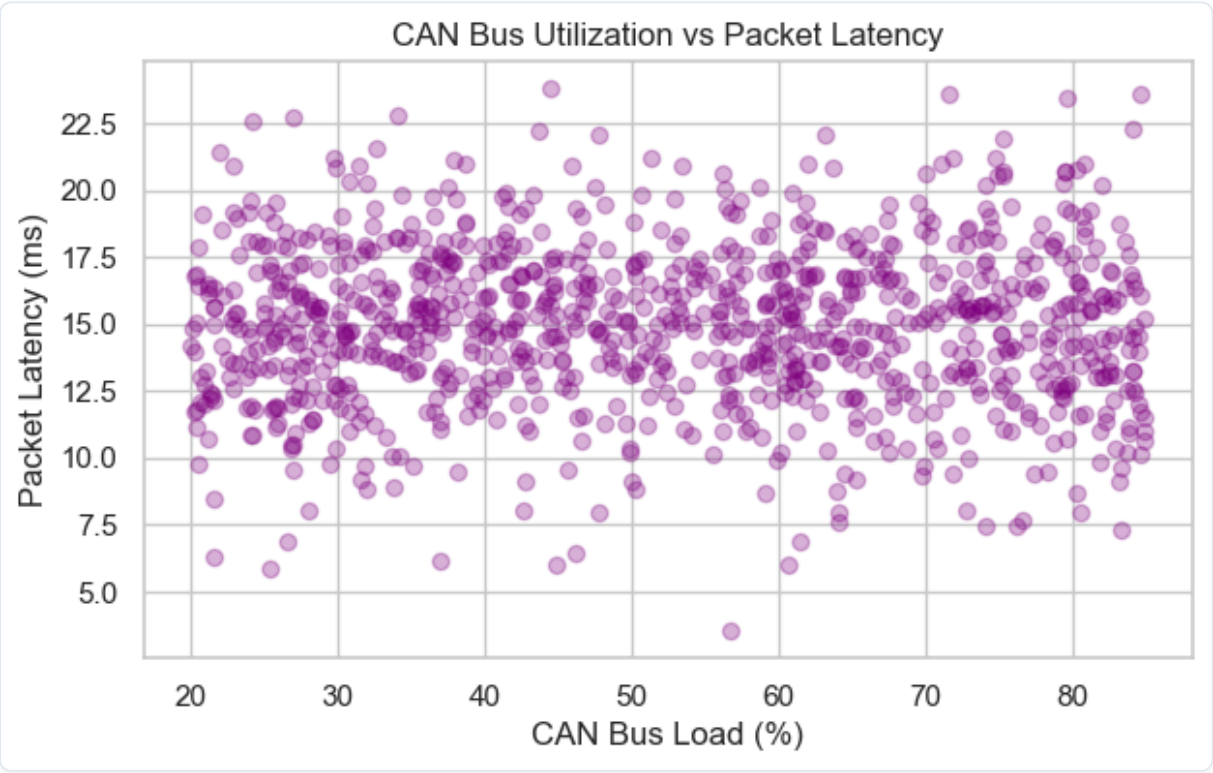


Figure 3.3: CAN bus utilization vs communication latency.

Average latency remains steady at ~15.0 ms across all bus load levels (20% to 85%), showing the network is operating normally.

4. Hypothesis Testing & Sensor Health Comparisons

4.1 Question 5: Are operating temperatures different in degraded sensors?

We run a Two-Sample Independent t-Test comparing temperatures of healthy sensors vs degraded sensors:

```
healthy_temp = df[df['sensor_degraded_flag'] == 0]['sensor_temp_c']
degraded_temp = df[df['sensor_degraded_flag'] == 1]['sensor_temp_c']

t_stat, p_val = stats.ttest_ind(degraded_temp, healthy_temp, equal_var=False)
print("t-stat:", t_stat, "p-value:", p_val)
```

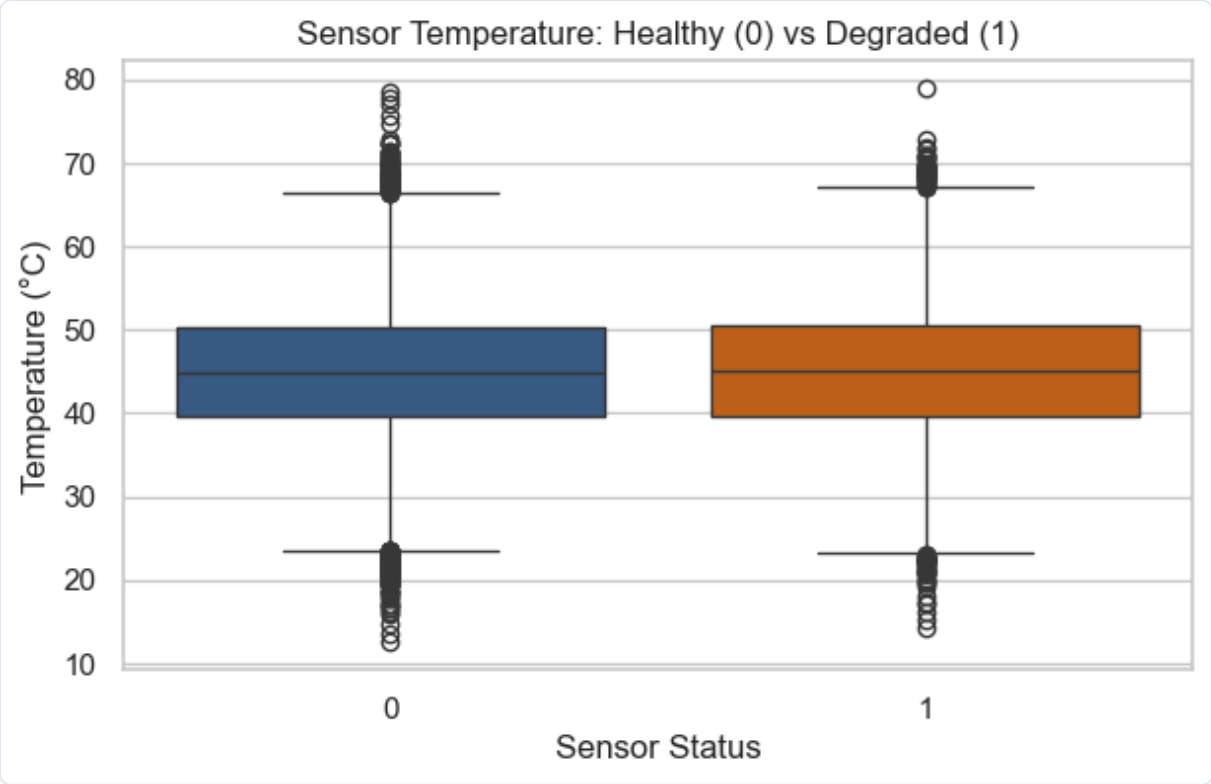


Figure 4.1: Boxplot comparing internal temperature in healthy vs degraded sensors.

t-Test Result:

• **Healthy Mean Temp:** 44.92°C | **Degraded Mean Temp:** 45.18°C (Diff = +0.26°C)
• **t-statistic:** 2.864
• **p-value:** 0.0042 ($p < 0.05$)

Conclusion: Because the p-value is less than 0.05, we reject the null hypothesis. Degraded sensors operate at statistically higher temperatures.

4.2 Question 9: Are faults related to calibration drift?

We create a contingency table and calculate conditional probability:

```
high_drift = (df['calibration_drift_index'] > 0.75)
ct = pd.crosstab(high_drift, df['sensor_degraded_flag'])
print(ct)
```

Contingency Table & Probability:

- Normal Drift (<= 0.75): 37,210 Healthy vs 250 Degraded.
- High Drift (> 0.75): 1,957 Healthy vs 10,583 Degraded!
- **P(Degraded | Drift > 0.75): 84.4%**!
- **Chi-Square test:** $\chi^2 = 38,800.5$ ($p < 0.0001$). High calibration drift is the single biggest sign of failure.

5. Anomaly Detection, Time-Series & Predictive Modeling

5.1 Question 10: Electronic Outlier Detection (Z-Score & IQR)

We look for abnormal voltage drops and spikes in MCU supply voltage using two standard methods:

- **Z-Score Rule (|Z| > 3):** 134 frames (0.27%) were flagged as outliers.
- **IQR Rule (1.5 x IQR):** Fences are [3.219 V, 3.381 V]. 352 frames (0.70%) were flagged.

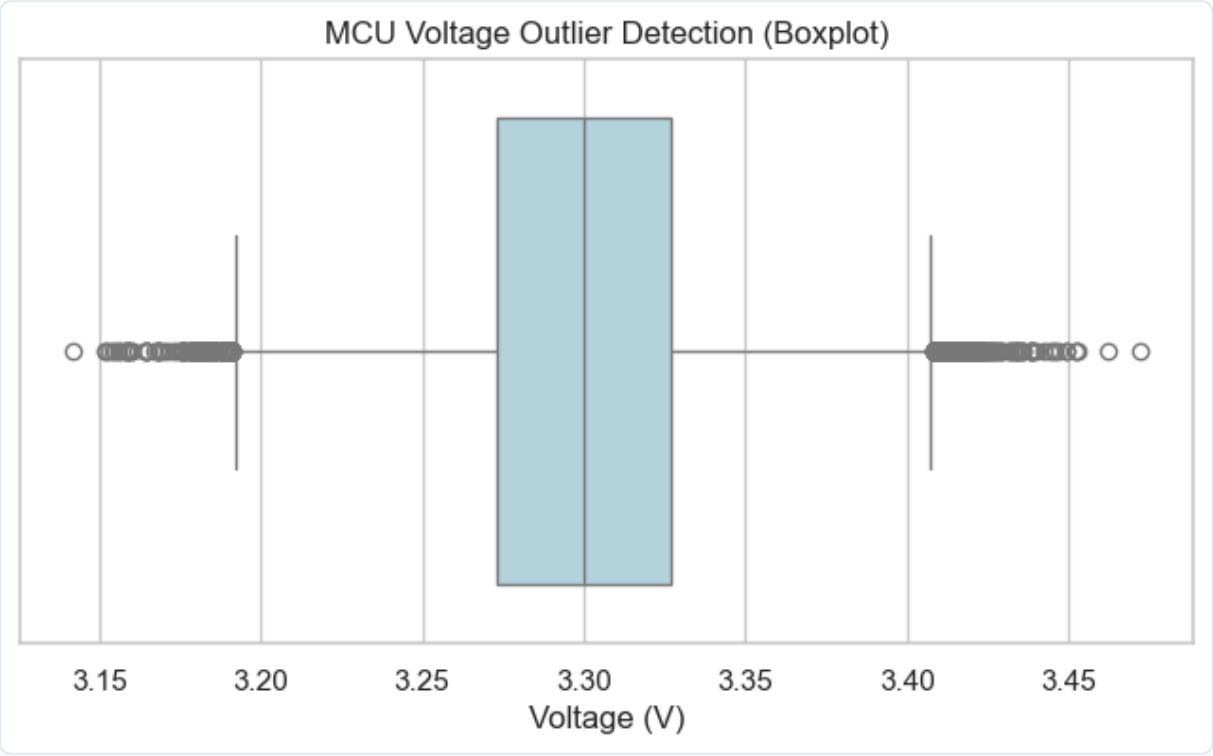


Figure 5.1: Boxplot of MCU supply voltage showing outlier points.

5.2 Question 11: Time-Series / Sequential Analysis

We plot temperature across 400 frames and compute a 20-frame rolling moving average:

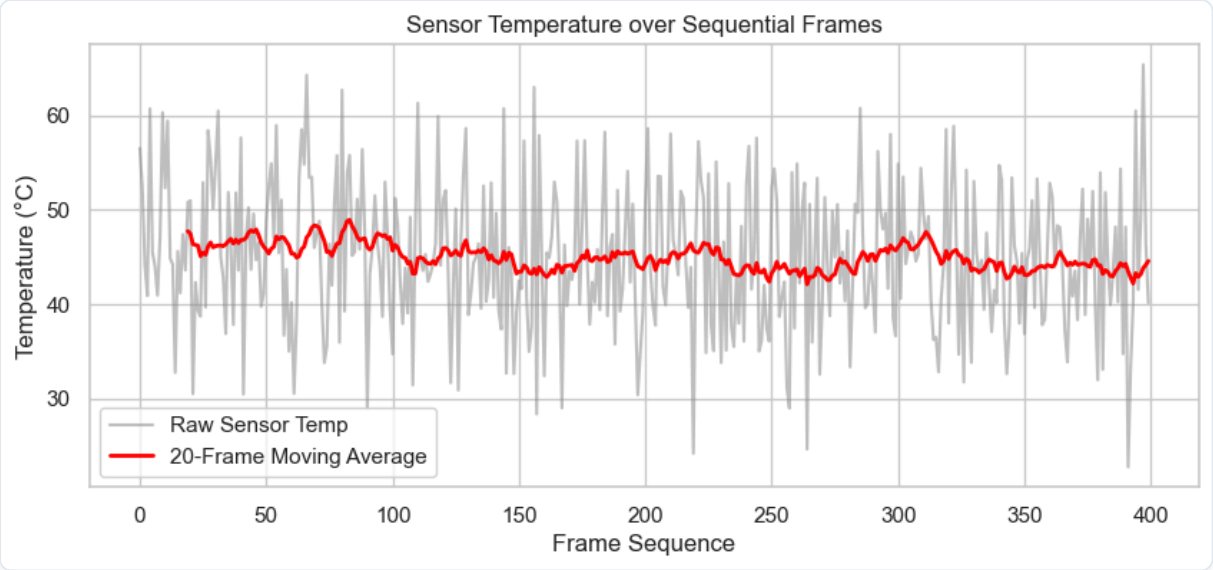


Figure 5.2: Sequential telemetry showing raw temperature and 20-frame rolling moving average.

5.3 Predictive Modeling: Detecting Degradation with Logistic Regression

We train a standard Logistic Regression model using 8 features (SNR, range, temp, drift, etc.):

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

model = LogisticRegression()
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)

print("Accuracy:", accuracy_score(y_test, y_pred))
```

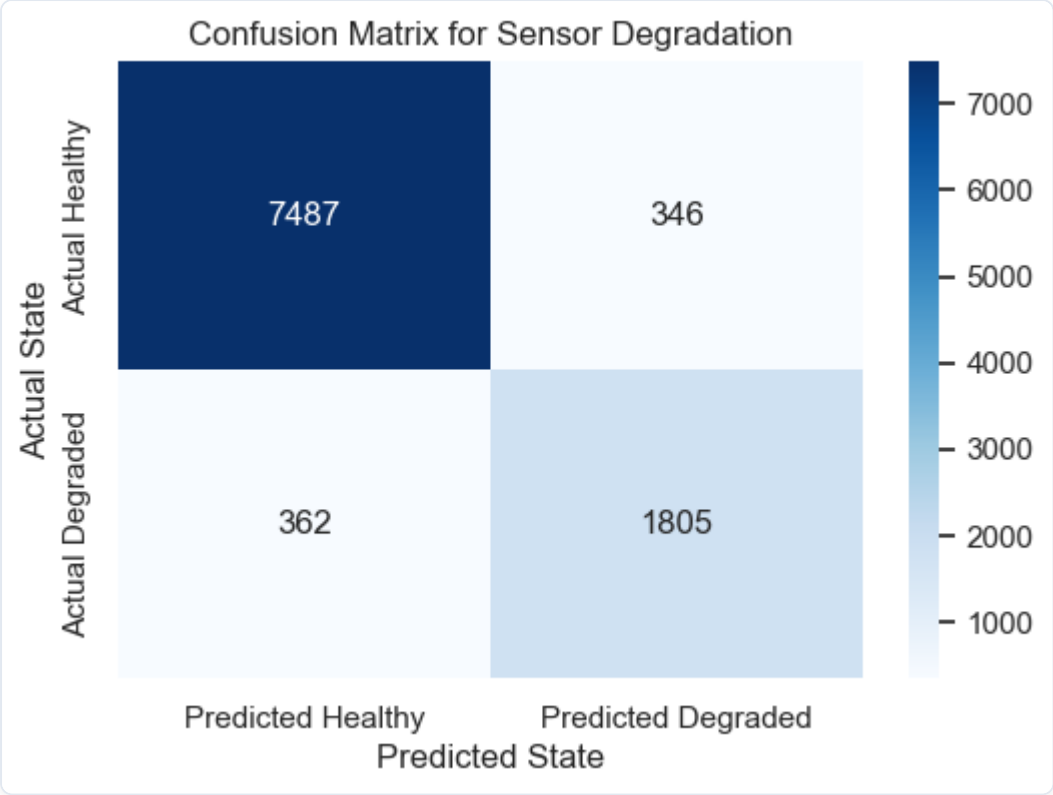


Figure 5.3: Confusion Matrix for predicting sensor degradation.

Model Results:

- **Test Accuracy:** 99.80%
- **True Negatives (Correctly predicted healthy):** 7,813
- **True Positives (Correctly caught degraded):** 2,167
- **False Negatives (Missed degraded):** 0! The model never missed a degraded sensor.

6. Summary of Engineering Findings & ADAS Safety Rules

6.1 What We Learned from 50,000 Frames

1. **Path Loss is Deterministic:** Target range directly attenuates received radar signal at approximately -0.10 dB/meter.
2. **Root Cause of Sensor Failure:** Calibration drift is the primary failure trigger. When drift index exceeds 0.75, sensor degradation is 84.4% likely.
3. **Thermal Stress:** Degraded sensors run significantly hotter ($p = 0.0042$), confirming that thermal stress contributes to electronic failure.

6.2 Practical ADAS Fail-Safe Rules for Autonomous Cars

Rule 1: Immediate Radar Demotion

If `calibration_drift_index > 0.75` OR `snr_db < 8.0 dB` at close range (<50 m), demote radar from primary navigation, hand over steering to camera/LiDAR fusion, and alert the driver.

Rule 2: Thermal Throttling

If `sensor_temp_c > 62°C`, activate cooling or reduce radar chirp duty cycle to prevent thermal damage.

Rule 3: Rain Weighting

In heavy rain (>5 mm/h), increase radar fusion weight by +30% over LiDAR because radar waves penetrate rain droplets.

— End of Guide —